11.Diffie Hellman Key Exchange

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<!DOCTYPE html>
<html>
<head>
   <title>Diffie-Hellman Key Exchange</title>
</head>

<body>

<h2>Diffie-Hellman Key Exchange</h2>

<label>Prime Number (p):</label>
<input type="number" id="p" >
<br><br>

<label>Primitive Root (g):</label>
<input type="number" id="g" >
<br><br>

<label>Alice's Private Key (a):</label>
<input type="number" id="a" >
<br><br>

<label>Bob's Private Key (b):</label>
<input type="number" id="b" >
<br><br>
```

```
<button onclick="calculateKey()">Calculate Keys</button>

<h3>Result</h3>

<p id="alicePublic"></p>
<p id="bobPublic"></p>
<p id="aliceSecret"></p>
<p id="bobSecret"></p>
<p id="message"></p>

<script>
function calculateKey() {
    let p = Number(document.getElementById("p").value);
    let g = Number(document.getElementById("g").value);
    let a = Number(document.getElementById("a").value);
    let b = Number(document.getElementById("b").value);

    // Calculate public keys
    let alicePublic = Math.pow(g, a) % p;
    let bobPublic = Math.pow(g, b) % p;

    // Calculate secret keys
    let aliceSecret = Math.pow(bobPublic, a) % p;
    let bobSecret = Math.pow(alicePublic, b) % p;

    document.getElementById("alicePublic").innerHTML =
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        "Alice's Public Key: " + alicePublic;

    document.getElementById("bobPublic").innerHTML =
        "Bob's Public Key: " + bobPublic;

    document.getElementById("aliceSecret").innerHTML =
        "Secret Key calculated by Alice: " + aliceSecret;

    document.getElementById("bobSecret").innerHTML =
        "Secret Key calculated by Bob: " + bobSecret;

    if (aliceSecret == bobSecret) {
        document.getElementById("message").innerHTML =
            "<b>Key exchange successful.</b>";
    } else {
        document.getElementById("message").innerHTML =
            "<b>Key exchange failed.</b>";
    }
}
</script>

</body>
</html>
```